	FACULTY OF COMPUTING		MARKS:
UNIVERSITI MALAYSIA PAHANG AL-SULTAN ABDULLAH	COURSE: Web Engineering	CODE: BCS2243	
	TOPIC: Chapter 1-9	SEM: 2 2025/2026	
	ASSESSMENT: Project	DURATION: 13 weeks	

General Instructions:

1. Each group can have a maximum of 4 members.
2. Please refer:
   (a) Table 2 for the project activities and datelines of submission.
   (b) Appendix 1 for the project description.
   (c) Appendix 2 for the rubric evaluation.
   (d) Appendix 3 for the cover page for each report submission.
3. <u>Any late submission or plagiarism will be penalized.</u>

Course Outcome:

The Course Outcome (CO) for this assessment as followed:

Course outcome	Outcome description
CO1	Design appropriate solution using fundamental web engineering concepts.
CO2	Construct a web-based application prototype using HTML, web server, database and scripting language based on web engineering methodology.
CO3	Demonstrate communication effectively in written and oral form through group discussion, meeting and presentation session.

The CO and marks distribution is summarised in Table 1:

Table 1. Assessment percentage according to CO

Assessment type		Course outcome percentage			
		CO1	CO2	CO3	Total (%)
Proposal		5	5	0	10
Progress	Progress 1	0	2	0	5
	Progress 2	0	3	0	
Project	Prototype	7	8	0	15
	Presentation	0	0	10	10
			Grand total (%)		40

Question:

You are required to develop a web application prototype. The requirement of the application is attached in the Appendix 1. Table 2 summarised the project activities and submission.

Table 2. Project activities and submission dateline

Week	Day/Time	Assessment percentage	Milestones
8	8 May 2026	10%	Project proposal submission
9	Lab session C1: 12 May 2026 C2: 11 May 2026 C3: 14 May 2026 (Changes will be announced by the lecturer)	2%	Progress presentation 1: Development environment setup Database created Web page design Login module
10	Lab session C1: 19 May 2026 C2: 18 May 2026 C3: 21 May 2026 (Changes will be announced by the lecturer)	3%	Progress presentation 2: User session Manage data (insert, update, delete, view)
13	Lecture and Lab session 8 – 12 June 2026 (Presentation date and time will be announced by the lecturer)	25%	Complete prototype from deployment server Final report submission

Students are required to prepare a proposal consists of the following items. The detailed evaluation is attached in **Appendix 2**.

1. The idea / principle of the solution
2. Review of **TWO EXISTING SYSTEMS** that similar to your idea
3. Project Management Framework
4. Project Requirements
5. Proposed Design (diagram required: use case diagram, use case scenarios, hypertext structure model and access model)
6. Data Design (Entity Relationship Diagram (up to 3NF) and data dictionary)
7. Presentation model for each module (Graphical user interface design)
8. Testing Plan
9. Deployment Plan

Appendix 1

Project name: **FK Student Club & Event Management System**

The Student Affairs Department at Faculty of Computing, Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA) which located in Pekan is having a main problem encountered by the administration in effectively monitoring student organizations and campus activities. Your development team is assigned to propose and design a web-based solution which help in managing student club profiles, streamlining events and programs and tracking student attendance for extracurricular activities across the faculty. You are required to hold progress meetings with stakeholders every three weeks until the application is delivered. The basic requirements for the project are:

Module 1: Login, Manage Membership and User Registration

1. *Login*

The login page with a web session is designed for different types of users: administrator (FK staff), club committee members and registered students (undergraduate and postgraduate). Each user is required to log in using their registered username and password. After successful authentication, users will be redirected to their respective dashboards based on their role. The system will display the registered user's name and role on the main interface to indicate that the login session is active.

2. *Manage membership*

All students must have a registered account before accessing the application. User registration will be performed by the system administrator. During the registration process, the administrator will insert user details such as student ID, name, email address and role. Once registered, students will be able to join or participate in student clubs available in the system. Club committee members will also be registered under specific clubs and assigned roles such as president, secretary, treasurer or committee member. The system ensures that only authorized users can manage club-related activities.

3. *User profile management*

This function allows users to manage their personal information within the system. Users can view, update, or edit their profiles, including contact details, profile photo, and club membership information. The system administrator is authorized to create new users, update existing user profiles, delete inactive accounts, and view all registered user information. This functionality ensures that the user database remains updated and accurate.

4. *Administrator dashboard*

The student responsible for this module is also required to develop the Administrator Dashboard. The dashboard will present summarized information related to users and club activities such as:

- Total number of registered students
- Total number of active clubs
- Number of upcoming events
- Recent user registrations

The dashboard should display information in graphical formats such as bar charts, pie charts or line graphs to assist administrators in monitoring user participation and system usage.

Module 2: Manage Student Clubs

1. *Manage club information*

The administrator is responsible for managing all student clubs within the faculty. This includes creating, updating, deleting and viewing club details. Each club record will contain information such as club name, description, advisor name and committee members. The administrator may also activate or deactivate clubs depending on their operational status. This ensures that only active clubs are visible to students in the system.

2. *Manage club committees*

Each club must have a set of committee members responsible for managing club activities. The system allows administrators to assign or update committee roles, including positions such as president, vice president, secretary and treasurer. Committee members will have additional privileges within the system, such as managing events and viewing event participation records related to their respective clubs.

3. *View club list and details*

All users will be able to view the list of available clubs within the Faculty of Computing. Students can browse club information including club description, committee members, past and upcoming events. This feature helps students discover clubs and stay informed about activities organized by different student organizations.

4. *Administrator dashboard*

The student responsible for this module is also required to develop the Club Management Dashboard. The dashboard should present summarized information such as total number of clubs in the faculty, number of active clubs, number of students involved in clubs and distribution of students across clubs. These statistics will be presented using graphs and charts to assist administrators in understanding student involvement in extracurricular activities.

Module 3: Manage Events and Event Registration

1. *Create and manage events*

Club committee members are responsible for organizing events related to their respective clubs. The system allows committees to create, update, delete and view event details. Each event record will include information such as event title, event description, event date and time, venue or location and maximum number of participants. Once an event is created, it will be visible to all registered students through the system.

2. *Event registration*

Students can browse upcoming events and register for events that interest them. The system will allow students to register, cancel, update and view their event registration history. Students must confirm their registration details before submitting the registration form. Once registration is successful, the system will record the participant information in the event participant list.

3. *Registration validation and capacity control*

The system will automatically check for event capacity limitations to prevent over-registration. If the maximum number of participants has been reached, the system will notify students that the event is fully booked. The system may also allow event organizers to create a waiting list for students who wish to participate if additional slots become available.

4. *Administrator dashboard*

The student responsible for this module will collaborate with the student responsible for **Module 4** to develop the Event Management Dashboard. The dashboard will present insights such as number of events organized by each club, number of participants for each event, popular events based on registration count and monthly event activity trends. The information should be displayed in graphical formats such as bar charts and line graphs.

Module 4: Manage Event Attendance and Participation Reports

1. *Manage event attendance*

During the event, committee members will record the attendance of registered participants. The system may use QR code scanning or digital check-in methods to confirm student attendance. Each attendance record will include information such as student name and ID, event name and date and time of attendance and attendance status (*present, absent, late*). This feature ensures accurate tracking of student participation in club activities. The system will automatically assign points for attendance based on the student's participation (refer Table A):

Table A. The enforcement according to accumulated point

Total point	Enforcement type
Present on time	+10 points
Late arrival	+5 points
Absent without notice	-10 points
Volunteer/helper in the event	+5 points

2. *Participation tracking and history*

The system will maintain a comprehensive history of student participation in events. The system allowing students to view their total points, participation history, and ranking. Meanwhile club committees and administrators can access these records to monitor engagement trends. Total points are automatically updated for each student after every event and the system uses the accumulated points to determine student recognition levels, such as active participant, highly active, or outstanding participant, providing a transparent and systematic way to evaluate and reward student involvement in club activities.

3. *Event reports and statistics*

The system will generate dynamic reports related to participation and points. These reports may include number of participants per event, attendance rate for each event, points accumulated by each student per event and overall semester and most active clubs based on event organization. These reports assist administrators in evaluating the effectiveness of student club activities.

Table B. Point-Based Evaluation

Total point	Student Recognition Enforcement
Less than 20	Warning / Reminder to participate more
20–49	Eligible for participation certificate
50–79	Eligible for active student award / bonus points
80 and above	Outstanding participant; eligible for leadership award / priority in event registration

4. *Administrator dashboard*

The student responsible for this module will also develop the participation and attendance dashboard. The dashboard will present summarized information such as total number of events conducted, total student participation in events, attendance rate per clubs and per events, list of most active students and club and graphical representation of points distribution, participation trends and engagement rankings. The information will be presented using visual analytics such as graphs and charts to support better decision-making for faculty administrators. The dashboard will calculate total points per student automatically, showing real-time recognition status. Administrators can filter by club, semester, or event type to identify top participants and committees.

General guidelines for project development and presentation:

1. Each group member is responsible for a specific module and required to:
    a. propose a detailed process
    b. develop the CRUD functionality (create, read, update and delete)
    c. develop appropriate reporting for the module. The report must include a single table and a join table using Structured Query Language (SQL) and they must be distinct from each other. Repetition of calculations and report features will result in a deduction of marks.
2. All modules must be integrated with one another and presented as a complete web application.
3. The final presentation (complete prototype) is required to be hosted on Indah server or any free web hosting platform.
4. Absence from the final presentation will result in a zero mark.
5. You are recommended for a backup strategy by uploading your documents to online storage options (such as OneDrive, Google Drive, email or free web hosting).
6. A marks penalty will be applied for late submissions, while a score of zero will be given for failure to submit.

Appendix 2: Rubric Evaluation

2.1 Peer Review

Peer review will be evaluated using a google form make available to you in KALAM upon your submission date. The evaluation items for peer review are listed in Table 1. Each of the group member must fill in peer review form once it is available in KALAM.

Table 1. BCS2433 WEB ENGINEERING PROJECT: PEER EVALUATION FORM SEM 2 2025/2026

Course Learning Outcomes:	
CLO3: Work effectively in group and promote leadership's	PLO4: Communication Skills

PEER EVALUATION RUBRICS						
ITEM		SCORE				
		0	1	2	3	4
A.	CONTRIBUTION	Very Low	Low	Moderate	High	Very High
	Contribute to the effort in writing as well as in team	Your partner had contributed 0%- 19% of the report writing and and team effort (eg: absent in every meeting @ attend meeting but quiet) - sleeping partner	Your partner had contributed 20%- 49% of the report writing and and team effort (eg: partially attend and team has to do for him/her part)	Your partner had contributed 50%- 69% of the report writing and and team effort (eg: attend meeting but team has to cover his/her part)	Your partner had contributed 70%- 89% of the report writing and and team effort (eg: attend all meetings but his/her report was submitted late)	Your partner had contributed 90%- 100% of the report writing and and team effort (eg: full attendance in every meeting and complete his/her part in report)
B.	PARTICIPATION	Very Passive	Passive	Normal	Active	Very Active
	Participate actively in leading / facilitating discussion / team meetings / developing ideas and planning project (every meeting must provide logbook and snap group photo - as an evidence)	Your partner did not participate at all (eg: empty record in logbook)	Your partner participated above 20% of all activities in group	Your partner participated in between passive and active. Sometimes passive and sometimes active. (50% attended)	Your partner participated actively in most activities	Your partner participated actively in all activities in your group
C.	COLLABORATION	Entry Level	Adoption Level	Adaptation Level	Infusion Level	Transformation

	Worked cooperatively with other group members regards races, genders and seniority	Your partner was unable to collaborate in your group although he/she exists.	Your partner had taken action to collaborate but not in depth	Your partner had certain blending with all team members between the given expectation	Your partner had ability to mixture and collaborate with all team members	Your partner had transformed not only him / herself but also other member in class
D.	ATTITUDE	Very Negative	Negative	Neutral	Positive	Very Positive

	Displayed positive approach and made constructive comments in working toward goal	Your partner showed extreme negative feedback (eg: not interested to complete his/her tasks and create hatred and sabotage your	Your partner showed negative feedback (eg: not interested to complete his/her tasks only)	Your partner had neutral feedback which did not showed any extreme positive or negative actions	Your partner has shown positive feedback in his/her actions in team	Your partner had shown extremely positive not even in his/her group but also in overall member in the class
PEER EVALUATION RUBRICS						
ITEM		SCORE				
		0	1	2	3	4
E	COMMUNICATION	Never	Rarely	Sometimes	Frequent	Always
	Expressed thoughts clearly and show efforts in communicating well within group	our partner was blurred in all discussions	Your partner was unable to give clear explanations and sometime can talk but the fact is not true	Your partner was able to give some parts of clear explanations	Your partner was able to deliver his / Her understanding within knowledge in teaching learning	Your partner was consistently deliver his communication with fluent and good knowledge (eg: the way he / she talk is truth and polite)
F	RESPONSIVENESS	Unresponsive	Responsive t	Responsive to	Awake	Alert
	Reacted sensitively to verbal and nonverbal cues of other team members within group	Your partner was unable to respond for any request in his / her tasks	Your partner was unable to respond verbal	Your partner was unable to respond nonverbal	Your partner was able to respond to verbal and nonverbal with good respond	Your partner was able to respond to verbal and nonverbal with advanced respond

USING THE ABOVE RUBRICS, EVALUATE YOUR TEAM MEMBER (By giving appropriate scores)

GROUP NAME:	EVALUATION CRITERION (BASED ON ABOVE RUBRIC RANK 0 TO 5)
YOUR NAME (STD ID):	
YOUR TEAM MEMBER:	

No	Name (Std ID)	A	B	C	D	E	F	TOTAL
1								
2								
3								
4								

Appendix 3: Sample Front Page Cover

UNIVERSITI MALAYSIA PAHANG AL-SULTAN ABDULLAH

FACULTY OF COMPUTING

BCS2243
WEB ENGINEERING

SEMESTER 2 2025/2026

TITLE : (Project Name)
SECTION : 01 / 02 (Remove unnecessary info)
LECTURER : (Your lecturer's name)

Student Detail: (Adjust all photos to fit in one page)

Name	Student ID	Student Photo

-END-